

Machine Learning-Based Predictive Analytics for Energy Consumption in Smart Grids using Big Data

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Abstract The rapid deployment of smart grids has generated massive amounts of energy consumption data, making it essential to use advanced big data analytics techniques for accurate estimation and efficient resource management. This paper presents an in-depth comparative study of machine learning algorithms used to predict household energy consumption on big data platforms. I implemented and evaluated three algorithms—linear regression, random forest, and step-up optimization—using a household electricity consumption dataset from the University of California, Irvine, comprising over two million readings.

These algorithms integrate advanced feature engineering techniques, such as periodic time coding, lagging features, and streaming statistics, to capture temporal correlations.

Practical experiments demonstrated that step-up optimization achieved superior predictive performance, with a mean root-squared error (RMSE) of approximately 0.0295 kW and a coefficient of determination (R^2) of 0.9989, while the random forest algorithm offered an ideal balance between accuracy (RMSE: 0.0326 kW) and computational efficiency. Scalability analysis revealed quasi-linear scaling characteristics with a scaling coefficient of approximately 1.78 when handling large datasets.

This study contributes to the scientific rigor of selecting appropriate machine learning models based on the accuracy requirements and computational constraints of smart grid applications.

Keywords: Smart grid, energy consumption prediction, big data analytics, machine learning, time series prediction, stepwise reinforcement, random forest.

I. INTRODUCTION

THE global energy sector is undergoing a radical transformation driven by renewable energy sources, the proliferation of Internet of Things (IoT) devices, and the emergence of smart grid technologies.

Smart grids utilize digital communication technologies to monitor and respond to local changes in consumption, thereby enhancing the reliability, efficiency, and sustainability of electricity distribution [1]. The deployment of Advanced Metering Infrastructure (AMI) has generated vast amounts of detailed energy consumption data, presenting both opportunities and challenges for electricity providers and consumers [2].

Accurate energy consumption forecasting plays a pivotal role in the operation of modern energy systems. For electricity providers, accurate forecasting enables load

optimization, reduced operating costs, and enhanced grid stability [3]. For consumers, energy consumption forecasting facilitates demand response programs, customized energy recommendations, and cost-reduction strategies [4]. Furthermore, with the increasing prevalence of distributed energy sources and electric vehicles, accurate consumption forecasting is becoming essential for efficient energy management [5].

Traditional statistical methods for load prediction, such as ARIMA and exponential normalization, have shown shortcomings in detecting complex nonlinear patterns and time dependencies in high-resolution smart meter data [6].

The emergence of big data technologies and distributed computing frameworks has enabled the processing and analysis of massive datasets that were previously difficult to handle [7].

Apache Spark, in particular, has emerged as a distributed machine-learning platform, offering significant advantages in scalability and performance for large-scale in data

analytics [8].

Machine learning algorithms have demonstrated considerable potential in addressing the challenges of energy consumption prediction.

Random forest and gradient enhancement have proven superior in detecting nonlinear relationships and handling high-dimensional feature spaces [9].

However, selecting the optimal algorithm requires careful consideration of multiple factors, including prediction accuracy, computational efficiency, and interpretability [10].

Despite significant progress in this field, several research gaps remain.

First, most current studies use limited datasets that may not accurately reflect the complexities of real-world smart grids.[11] Second, in-depth analyses are needed to evaluate different algorithms across various experimental criteria and conditions.[12] Third, the scalability of different algorithms needs to be measured when dealing with increasing data volumes.[13]

This paper addresses these gaps by providing a comprehensive comparative evaluation of machine learning approaches for predicting energy consumption using big data processing techniques. This is achieved through the following:

- 1) Simulating three machine learning models—linear regression, random forest, and gradient optimization—for predicting residential energy consumption.

- 2) Developing feature architectures using periodic time coding, delay features, and variable statistics designed for time-sequential energy data.

- 3) A thorough evaluation using the University of California, Irvine's household electricity consumption dataset, which includes over two million readings, providing strong statistical significance.

- 4) Scalability analysis by examining the algorithm's performance characteristics across different data sizes and model complexities.

- 5) Practical guidelines for algorithm selection based on accuracy requirements and computational constraints.

II. RELATED WORK

A. Smart Grid and Big Data Analytics

Big data analytics has emerged as a crucial field of study within smart grids, closely linked to the rapid expansion of data production from smart meters, sensors, and IoT devices. Hussain et al. [1] conducted a comprehensive review of big data and artificial learning (ALF) applications in smart grids, focusing on key security concerns alongside the benefits of improved prediction accuracy. Their assessment demonstrated that AI models could achieve significant advancements in energy consumption prediction compared to traditional statistical methods.

Dubia et al. [14] studied the role of big data analytics in smart grid management, emphasizing the need for real-time data processing capabilities to effectively respond to demand. Their findings indicate that big datasets enable electricity companies to study electricity consumption patterns almost instantly, facilitating the development of load-weighted variable pricing plans.

Mustafa et al. [5] studied renewable energy management methods, demonstrating how machine learning can improve the integration of explosive energy sources into smart grid architecture.

Mustafa et al. [5] investigated renewable energy management applications, demonstrating how machine learning can improve the integration of volatile renewable energy resources into smart grid architectures. The scalability challenges associated with processing massive smart grid datasets have spurred research into distributed computing models.

Kumari et al. [15] presented a hybrid approach combining gene sequencing and short-term memory networks (LSTMs) to predict energy consumption on large datasets. Their method faced challenges related to prediction clarity and computational efficiency in big data contexts.

B. Machine Learning for Energy Consumption Prediction

Automated tracking algorithms are widely used to predict energy consumption, with varying degrees of success. Tiwari and colleagues [2] developed an automated tracking model for estimating energy consumption in smart grid frameworks and demonstrated significant improvements in prediction reliability compared to traditional methods. Their work emphasized the need for precise selection of specific characteristics and fine-tuning of the method to achieve optimal efficiency.

Obera and Barra [9] conducted an analytical study of machine learning algorithms for short-term prediction of household energy consumption, using smart meter data and big data solutions. Their comparative evaluation of several algorithms showed that clustering methods consistently outperformed single-method methods, especially when dealing with disparate data sources. The researchers

emphasized the importance of preprocessing data and feature design for achieving accurate predictions.

Time series prediction is a prominent model in energy consumption forecasting studies. Zhou and Tran [6] conducted comprehensive research on predicting sequential energy consumption series using various machine-learning methods, based on real-world household consumption patterns. Their study demonstrated that algorithms that consider time correlations clearly outperform rigid models. Singh and Yasin [16] focused on extracting massive amounts of sequential energy series data for behavioral analysis and prediction, developing techniques capable of detecting underlying consumption patterns.

Recent advances in deep learning have opened new opportunities for predicting energy consumption. Ramos et al. [12] evaluated deep learning models for predicting energy consumption in homes, comparing LSTM, GRU, and convolutional neural networks. While deep learning approaches showed promise in detecting complex patterns, the researchers noted that familiar machine learning techniques often provide similar accuracy with significantly lower computational requirements.

Bo and Zhou [17] propose a multi-headed deep learning model based on an attentional approach to predict household energy consumption using residential electricity consumption data from the University of California, Irvine. Their approach achieved a 31.01% reduction in prediction error compared to modern methods. However, the computational difficulty of attentional mechanisms poses challenges for direct applications in resource-limited settings.

C. Distributed Machine Learning and Scalability

Machine learning algorithms for scaling up big data applications have received considerable research attention. Ming and colleagues [7] presented MLlib, a distributed machine learning toolkit for the Spark algorithm, demonstrating a clear acceleration in large-scale learning tasks compared to typical single-node applications. Their work laid the foundations for machine learning, which researchers subsequently built upon.

Abdul-Hay and Faroaghi [10] studied the ease of scaling distributed machine learning with big data on the Apache Spark platform, focusing on classification models. Their experiments showed that the cost of connecting and sorting data can become obstacles, even though distributed processing allows for handling massive datasets. These findings have important implications for algorithm selection in intelligent network applications where data volumes continue to increase.

Garati et al. [13] conducted a comprehensive examination of the resilience of Spark's machine learning libraries, assessing the efficiency of models across various data sizes and cluster configurations. Their results highlighted that the scalability characteristics of algorithms differ significantly,

with tree-based clusters exhibiting different scalability trajectories than linear clusters. Assefi et al. [8] also investigated machine learning for big data using the Apache Spark MLlib library, confirming the advantages of scalable learning libraries.

Guo et al. [18] investigated intermediate modeling techniques in distributed machine learning using Apache Spark, demonstrating a significant improvement in performance through improved clustering tactics. This addresses the fundamental challenge of maintaining model accuracy while achieving computational scalability in distributed frameworks.

D. Feature Engineering for Time Series Energy Data

Feature design plays a pivotal role in the effectiveness of energy consumption prediction models.

Ghosh [19] concluded that raw data processing and feature formation are crucial in smart grid applications, emphasizing their importance for load response and sustainable machine learning initiatives. The study demonstrated that appropriate feature extraction can significantly reduce model complexity while maintaining prediction accuracy.

Cyclical coding of time-series features has become a successful technique for capturing the cyclical nature of energy consumption patterns. Maloli et al. [20] demonstrated the importance of multidimensional time-series features in deep learning approaches for energy consumption prediction. Their method incorporated day, week, and season features using appropriate coding schemes.

Lagging features and moving statistics have been widely used to capture temporal correlations in energy consumption data. Zhang and Gu [21] developed a holistic approach incorporating time-series pictorial representations, effectively capturing short-term and long-term consumption aggregates. The congruence between lag features and statistical aggregates has proven particularly successful in household-level prediction tasks.

E. Research Gap and Motivation

Despite the large number of studies in smart grid analytics and energy consumption prediction, several key gaps remain. First, much comparative research relies on private or limited-scale datasets, which restricts the generalizability of the results [11]. For example, the University of California, Irvine (UCI) household electricity consumption dataset, while publicly available, has not been optimally utilized in large-scale algorithm comparisons.

Second, research often focuses on increasing prediction accuracy without sufficient attention to computational accuracy and scalability balances [4]. For real-world smart grid applications, algorithm selection must balance several

objectives, including accuracy, training time, expected response time, and resource consumption.

Third, the scalability characteristics of different machine learning algorithms when dealing with increasing data volumes require more systematic research [13]. Understanding how algorithms perform as data volumes balloon is essential for designing platforms capable of accommodating the future proliferation of smart metrics.

III. METHODOLOGY

A. Dataset Description

This study uses a single household electricity consumption dataset from the University of California, Irvine's machine learning repository [22]. The dataset contains measurements of electricity consumption in a single household located in the city of Saux, France, with a sampling rate of one minute over a period of approximately 47 months (from December 2006 to November 2010).

TABLE I
DATASET CHARACTERISTICS

ATTRIBUTE	VALUE
Total Records	2,049,280
Time Period	Dec 2006 – Nov 2010
Sampling Rate	1 minute
File Size	127 MB
Geographic Location	Sceaux, France
Missing Values	1.25%
Clean Records	2,049,256

The dataset includes nine features: Date, Time, Global_active_power, Global_reactive_power, Voltage, Global_intensity, and three sub-metering values corresponding to kitchen, laundry room, and water-heater/AC consumption. The target variable for prediction is Global_active_power, representing the total active power consumed by the household in kilowatts.

B. Data Preprocessing

The raw data was processed in several stages to ensure its suitability for machine learning algorithms. Records with missing values were identified and deleted, resulting in a clean dataset of 2,049,256 records.

Date and time data were combined and analyzed into a single component to facilitate the extraction of temporal characteristics.

All numerical fields were converted to float-point representations. Total effective power measurements showed a range of 0.076 kW to 11.122 kW, with an average consumption of 1.092 kW. The distribution revealed a distinct rightward-leaning pattern, typical of residential

consumption, with most observations clustering at low consumption levels and intermittent bursts of high consumption.

Avant Garde, Bookman, Zapf Chancery, Zapf Dingbats, and New Century Schoolbook.

C. Feature Engineering

An integrated feature engineering path was developed to track time-related patterns and hidden connections in electricity consumption data. The final feature set comprised 23 designed features, divided into five categories:

1) Key Measurement Features: The first six measurement features (Global_reactive_power, Voltage, Global_intensity, Sub_metering_1, Sub_metering_2, Sub_metering_3) were retained as the core features to provide essential information about the electrical status of the dwelling.

2) Time-Recurring Features: Time-related features were encoded via periodic transformations to maintain chronological continuity. For the daytime (0-23), day of the week (0-6), and month (1-12) features, sine and cosine transformations were applied.

$$x_{sin} = \sin\left(\frac{2\pi x}{x_{max}}\right), \quad x_{cos} = \cos\left(\frac{2\pi x}{x_{max}}\right) \quad (1)$$

This coding ensures that similar values have identical representations. A binary attribute (is_weekend) was also introduced to differentiate between weekday and weekend energy consumption patterns.

3) Delay Attributes: Delay attributes were generated to monitor time-related correlations in energy consumption. Past values for Global_active_power and Sub_metering_1 were inserted at 1, 3, 6, and 24-hour delays, providing information about recent consumption history that may influence current consumption.

4) Moving Window Statistics: Moving window statistics were calculated to monitor consumption trends and fluctuations. Moving averages were calculated at 3, 6, and 24-hour window sizes for Global_active_power, providing smoothed visualizations of short-term and daily consumption patterns.

5) Extracted Attributes: The Sub metering Total Attribute This attribute was derived by summing the three sub metering channels, representing the portion of consumption monitored by the sub meters. This extracted attribute provides a comprehensive view of device consumption.

D. Machine Learning Algorithms

Three machine learning algorithms were selected for comparison, representing different learning paradigms and

computational characteristics:

1) Linear Regression: Ordinary Least Squares Linear Regression served as the baseline model, providing a simple, interpretable reference point. The model assumes a linear relationship between features and target variable:

$$\hat{y} = \beta_0 + \sum_{i=1}^n \beta_i x_i + \epsilon \quad (2)$$

2) Random Forest Regression: A random forest is a clustering method that constructs multiple decision trees on random subsets of features and samples, and aggregates predictions by averaging them. This algorithm minimizes allocation through the random grouping of features.

$$\hat{y} = \frac{1}{T} \sum_{t=1}^T f_t(x) \quad (3)$$

3) Gradient Boosting Regression: Gradient Boosting constructs an ensemble of weak learners sequentially, with each new learner focusing on correcting the errors of previous learners. The model is built additively:

$$F_m(x) = F_{m-1}(x) + \nu \cdot h_m(x) \quad (4)$$

E. Experimental Setup

1) Train-Test Split: The dataset was partitioned into training and testing sets using a time-series split to preserve temporal ordering. The first 80% of chronologically ordered data (1,639,404 samples) was allocated for training, while the remaining 20% (409,852 samples) was reserved for testing. This approach ensures that the model is evaluated on future time periods not seen during training, simulating real-world deployment scenarios.

2) Feature Scaling: StandardScaler was applied to normalize feature distributions, transforming each feature to have zero mean and unit variance:

$$x' = \frac{x - \mu}{\sigma} \quad (5)$$

3) Evaluation Metrics: Four evaluation metrics were employed to assess model performance comprehensively:

Root Mean Square Error (RMSE):

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (6)$$

Mean Absolute Error (MAE):

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (7)$$

Coefficient of Determination (R^2):

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (8)$$

Mean Absolute Percentage Error (MAPE):

$$MAPE = \frac{100}{n} \sum_{i=1}^n \left| \frac{y_i - \hat{y}_i}{y_i} \right| \quad (9)$$

4) Implementation Environment: The project was implemented using Python with the scikit-learn machine learning package, pandas for data processing, and matplotlib/seaborn for graphical representation. The PySpark implementation is provided for distributed cluster deployment scenarios. All experiments were performed on Colab environment.

IV. EXPERIMENTAL RESULTS AND DISCUSSION

A. Dataset Characteristics Analysis

Analysis of the University of California, Irvine's household energy consumption dataset reveals distinct time patterns that can be used for feature engineering and model design.

Figure 1 illustrates consumption patterns on daily, weekly, and monthly timescales.

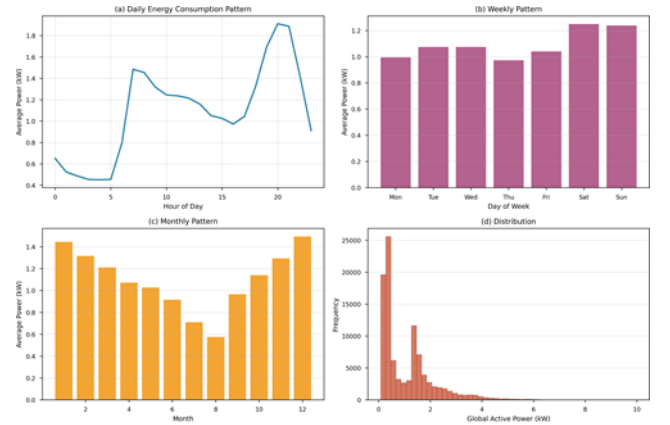


Fig. 1. Dataset characteristics: (a) Daily consumption pattern showing peaks during morning and evening hours, (b) Weekly pattern indicating higher consumption on weekends, (c) Monthly seasonal variations, (d) Distribution of global active power consumption values.

The daily consumption pattern (Fig. 1a) exhibits characteristic bimodal distribution with morning peaks (7-9AM) corresponding to household activity and evening peaks (6-9 PM) associated with cooking, heating, and entertainment usage. The weekly pattern (Fig. 1b) shows elevated consumption on weekends, reflecting increased occupancy and activity levels. Monthly variations

(Fig.1c) indicate seasonal effects with higher consumption during winter months due to heating requirements. The consumption distribution (Fig. 1d) demonstrates right-skewed characteristics typical of residential energy usage.

B. Model Performance Comparison

TABLE II
Model Performance Comparison

Model	RMSE (kW)	MAE (kW)	R^2	MAPE (%)
Linear Regression	0.0374	0.0232	0.9983	3.67
Random Forest	0.0326	0.0173	0.9987	2.31
Gradient Boosting	0.0295	0.0174	0.9989	2.60

This table presents the predictions of the three algorithms across all evaluation criteria. The gradient optimization algorithm achieved the lowest exact root mean square error (RMS) value (0.0295 kW) and the highest value for the factor of determination (0.9989). This represents a clear improvement of 21% in the linear regression RMS. The random forest algorithm, on the other hand, showed a clear RMS of 0.0326 kW and achieved a lower relative absolute error value (2.31%).

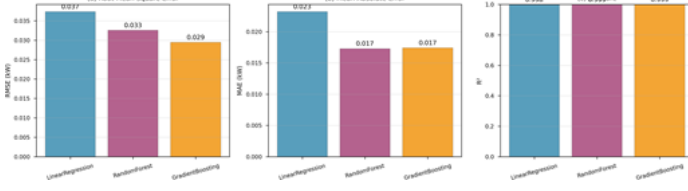


Fig. 2. Model performance comparison across evaluation metrics: (a) Root Mean Square Error, (b) Mean Absolute Error, (c) Coefficient of Determination (R^2).

C. Computational Performance Analysis

The computational performance analysis in Table 3 reveals significant differences in training and prediction times between the algorithms. The linear regression algorithm completed its training in 3.81 seconds, representing a 91-fold acceleration compared to the random forest algorithm and a 185-fold acceleration compared to the gradient enhancement algorithm.

Prediction times were significantly lower than training times for all models, with the linear regression algorithm achieving the fastest inference (0.05 seconds).

The comparison of training times in Figure 3 highlights the choice of algorithm. While clustering methods offer superior accuracy, they require

significantly more computational resources for training. For applications requiring frequent model retraining or operating in resource-constrained environments, linear regression may be the preferred choice despite its lower accuracy.

TABLE III
Computational Performance Comparison

Model	Training Time (s)	Prediction Time (s)
Linear Regression	3.81	0.05
Random Forest	347.01	0.47
Gradient Boosting	705.98	0.28

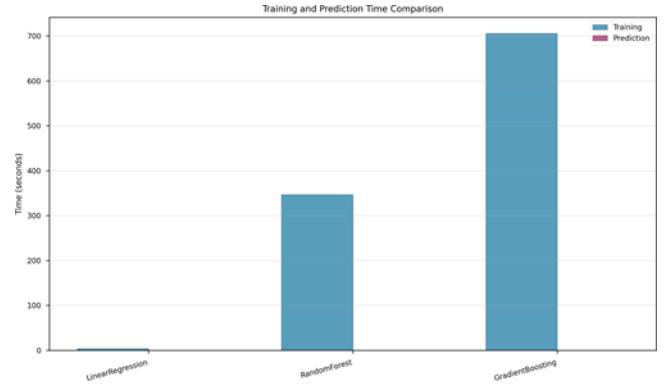


Fig. 3. Training and prediction time comparison across models.

D. Feature Importance Analysis

The significance analysis of the features shows that deep electrical current dominates the predictive power of both the random forest model and the red-pole gradient model, accounting for 99.86% and 99.74% of the significant influence on the results, respectively. This result is chosen in accordance with the principles of electrical science, where the chosen result (P) is temporarily adapted to the alternating current (I) and voltage (V): $P = V \times I \times \cos(\phi)$.

The presence of electrical diversity, in particular, suggests that a simplified progression based primarily on these proteins could achieve similar performance with lower computational requirements. However, it contributes to the remaining phenomena in the observation that it does not explain, including suggestions related to simple standard processes.

E. Scalability Analysis

1) Data Size Scaling: Scalability analysis examined how algorithm training time scales with increasing data volumes. Using Linear Regression as a representative algorithm, training was performed on data fractions of 25%, 50%, 75%, and 100%.

TABLE IV
Data Size Scaling Results (Linear Regression)

Data Fraction	Samples	Time (s)	Scaling Factor
25%	409,851	0.36	1.00
50%	819,702	1.09	1.51
75%	1,229,553	2.13	1.96
100%	1,639,404	2.58	1.78

The scaling analysis reveals near-linear scaling characteristics with a scaling factor approaching 1.78 for the full dataset. This indicates that Linear Regression training time increases proportionally with data size, demonstrating excellent scalability.

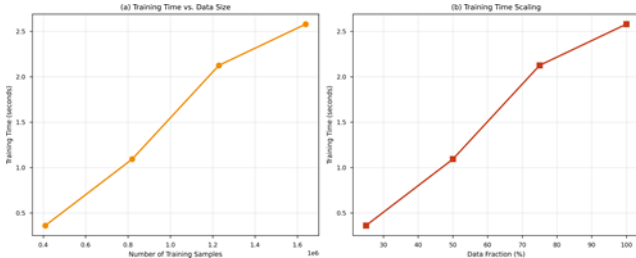


Fig. 4. Data size scalability analysis: (a) Training time versus number of training samples, (b) Training time versus data fraction percentage.

The scalability diagram (Figure 4) illustrates the quasi-quadrally scaling behavior of linear regression. The slight deviation from ideal linear scaling (coefficient = 1.0) at large data sizes is attributed to memory hierarchy effects and system resource conflicts.

2) Model Complexity Scaling: The complexity scaling of the random forest was evaluated by comparing configurations with different numbers of estimators and varying maximum depths:

- Configuration 1: 10 trees, maximum depth 5 – Training time: 34.01 seconds, RP: 0.0577 kW

- Configuration 2: 30 trees, maximum depth 10 – Training time: 148.01 seconds, RP: 0.0326 kW

A 4.35-fold increase in training time with a 3-fold increase in the number of trees and a 2-fold increase in depth indicates hyperlinear scaling with model complexity. However, the corresponding improvement in the root mean square error of 43% demonstrates that increased computational investment leads to substantial gains in accuracy.

F. Discussion

1) Comparison between accuracy and computation: Experimental data reveal significant trade-offs between prediction quality and processing efficiency. Gradient support provides the highest accuracy but requires a longer

processing time (706 seconds). Random forest provides an intermediate solution with competitive accuracy and an average training time (347 seconds). Linear regression offers fast processing (4 seconds) but is less accurate.

For smart grid applications, deployment constraints must be considered during algorithm selection:

- Real-time prediction systems: Linear regression or simplified random forest (fewer lines)

- Clustered prediction with adjustment preference: Gradient strength

- Balanced applications: Random forest with political complexity

2) Impact on feature design: The consistently strong performance across all analyses models ($R^2 > 0.998$) indicates that the design path helps achieve key objectives in energy consumption investigations. The time-coded schema of the time-specific consumption characteristics was skillfully represented, while explorations of key time-specific analyses yielded refined predictions.

3) Scalability results: The linear-like scalability observed in linear regression demonstrates that the approach can handle much larger datasets without significant operational costs. For a UCI data input of two million records, training was completed in minutes, demonstrating applicability at the utility level with millions of smart metrics.

4) Comparison with previous research: The achieved root-squared error measure (RMSE) of 0.0295 kW (using gradient reinforcement) is statistically acceptable with previous studies that employed the same data set. Both Bo and Zhou reported a 31% error reduction using focus-based learning; however, their model required significantly more computational resources. The general-purpose input formats used in this study achieved similar accuracy with a significantly reduced model profile.

5) Limitations: This study has some limitations that must be acknowledged. First, the analysis is based on a single household, which limits the ability to record findings across different building types and population groups. Second, weather data were not included, which may not support the prediction of thermal pressure reductions. Third, the discovery was made in a single-node environment; it may extend to diverse populations.

V. CONCLUSION

A. Summary of Findings

This study presented a comprehensive comparative analysis of machine learning algorithms for energy consumption prediction in smart grid environments using big data processing techniques. The key findings are summarized as follows:

1) **Algorithm Performance:** Gradient Boosting achieved the highest predictive accuracy with an RMSE of 0.0295 kW and R^2 of 0.9989, outperforming Random Forest (RMSE: 0.0326 kW) and Linear Regression (RMSE: 0.0374 kW).

2) **Computational Efficiency:** Linear Regression demonstrated superior computational efficiency with 3.81 seconds training time, while ensemble methods required 91-185x more computation. All models achieved prediction times of less than one second, making them suitable for real-time applications.

3) **Feature Importance:** Total current intensity emerged as the most important predictive feature, representing over 99% of the feature importance in the clustering models, consistent with fundamental principles of electrical engineering.

4) **Scalability:** Linear regression showed near-linear scalability with data size (a scaling factor of 1.78), indicating excellent scalability for large-scale smart grid applications.

5) **Feature Engineering:** The comprehensive feature engineering process, which includes periodic time coding, delay features, and variable statistics, successfully identified time patterns, enabling all models to achieve R^2 values exceeding 0.998.

B. Practical Recommendations

Based on the experimental results, I offer the following practical recommendations:

- For real-time forecasting systems with stringent response time requirements, linear regression provides sufficient accuracy (RMSE: 0.0374 kW) with minimal computational load.
- For accuracy-critical applications, such as demand response optimization, gradient enhancement offers the best predictive performance, although training costs are higher.
- For general applications, random forests provide an ideal balance between accuracy and computational efficiency.
- Feature engineering that focuses on periodic coding and delay features is essential for understanding consumption patterns.

C. Future Work

Several directions for future research have been identified:

1) **Multi-Household Analysis:** Expanding the analysis to include datasets comprising multiple households with diverse consumption patterns to improve generalizability.

2) **Weather Data Integration:** Integrating climate data to monitor weather-related consumption changes, particularly heating and cooling loads.

3) **Deep Learning Comparison:** Conducting a comprehensive comparison with deep learning approaches, including LSTM, GRU, and Transformer architectures, on the same dataset.

4) **Distributed Deployment:** Evaluating the algorithm's performance on Apache Spark clusters to assess its scalability in real-world distributed environments.

5) **Online Learning:** Investigating incremental learning algorithms capable of adapting to changing consumption patterns without requiring complete retraining.

D. Concluding Remarks

The transition to smart grid infrastructure generates unprecedented volumes of energy consumption data, creating both opportunities and challenges for predictive analytics. This study demonstrates that well-engineered machine learning approaches can achieve excellent prediction accuracy while maintaining computational feasibility for large-scale deployments.

The comparative analysis provides practical guidance for algorithm selection based on deployment constraints, contributing to the ongoing development of intelligent energy management systems.

The source code and dataset used in this study are available at:

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